Information Technology University



Lab Manual

for

CS220: Artificial Intelligence

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Lab-10

Lab Manual: Solving the 8 Queens Problem Using a Genetic Algorithm in Python

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Objective:

In this lab, you will implement a genetic algorithm to solve the 8 Queens problem. The goal is to place 8 queens on a chessboard such that no two queens threaten each other. You will use genetic algorithms to search for optimal solutions in Python.

Problem Overview:

The 8 Queens problem involves placing 8 queens on an 8x8 chessboard, ensuring no two queens threaten each other. You are required to use a genetic algorithm for this task, employing selection, crossover, and mutation to evolve solutions over generations.

Pre-requisites:

- Basic Python programming
- Familiarity with lists and functions in Python
- Basic understanding of genetic algorithms

Genetic Algorithm Overview:

- **Chromosome Representation**: Each chromosome represents a solution where the list of integers corresponds to the row position of the queen in each column.
- **Fitness Function**: Calculates how many queens are non-attacking.
- **Selection**: Select parents based on fitness.
- **Crossover**: Combine two parent chromosomes to create a child.
- **Mutation**: Occasionally mutate the child chromosome to introduce variation.

Useful Random Library Functions:

1. `random.randint(a, b)`
    - Generates a random integer between `a` and `b`, inclusive.
2. `random.choices(population, weights, k)`
    - Selects `k` elements from the population with replacement, where each element's probability is determined by the corresponding weight.
3. `random.random()`
    - Generates a random float between 0.0 and 1.0.

Lab Task:

1. **Chromosome Representation**: Implement a function to generate a chromosome representing queen positions on the chessboard.
2. **Fitness Calculation**: Write a function to calculate the fitness of a chromosome by evaluating the number of non-attacking queens.
3. **Selection**: Implement a method to select parent chromosomes based on fitness.
4. **Crossover and Mutation**: Implement functions for crossover between two parents and random mutation of chromosomes.
5. **Genetic Algorithm**: Use the above functions to build a genetic algorithm that evolves a solution for the 8 Queens problem.

Expected Output:

After running the algorithm, it should output the best solution and the number of generations it took to find it. The solution will be a chromosome representing the queens' positions.

Conclusion:

Reflect on the behavior of the genetic algorithm, including how many generations it took to find a solution and how effective the mutation rate and crossover mechanisms were.